

Homework 4

Math 3200—Fall 2026

Due 9/17 before class (turn in on gradescope)

Name:

1. Let $\sqrt{3}$ represent the unique positive real number whose square is equal to 3. Prove that $\sqrt{3}$ is irrational. The following outline might be helpful:
 - (a) Prove that if x is an integer that is not a multiple of 3, then x^2 is not a multiple of 3.
 - (b) Write the contrapositive of the statement proven in part (a).
 - (c) Assume for the sake of contradiction that $\sqrt{3}$ is rational. Use the statement you wrote in part (b) to prove something that simply cannot be true!

Use the principle of mathematical induction the following problems.

2. For all natural numbers n , show that

$$1^4 + 2^4 + 3^4 + \dots + n^4 = \frac{6n^5 + 15n^4 + 10n^3 - n}{30}.$$

3. Prove that $n! > 3^n$ for all $n \geq 7$.